

Compressed Data Structures for Query Processing

Advanced Topics in Data Analytics

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Kickoff meeting: 6 October 2026
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Compact storage and efficient queries

A sequence of one million zeros

00000000000000000000000000...

A compact description: *“repeat 0 one million times”*.

- Both descriptions represent exactly the same sequence.
- Can we read the value at a given position directly?
- Can we combine two sequences without expanding every element?

Selecting records by multiple conditions

Record	Country	Subscription
1	Austria	Premium
2	Germany	Basic
3	Austria	Basic
4	Italy	Premium
5	Austria	Premium
6	Germany	Premium

Which records match both conditions?

```
country = 'Austria' AND subscription = 'Premium'
```

Combining bitmap indexes with AND

One bit per record: **1** means that the condition holds.

Record	1	2	3	4	5	6
Austria	1	0	1	0	1	0
Premium	1	0	0	1	1	1
AND	1	0	0	0	1	0

Result

Records **1** and **5** match the query.

What if the table has a billion records?

One uncompressed bitmap

1,000,000,000 bits = 125,000,000 bytes = **125 MB**

- For country: one bitmap marks Austria, another marks Germany, and so on.
- With one bitmap per value, 1,000 distinct values require 1,000 bitmaps: **125 GB** of raw bit storage.
- AND/OR on uncompressed bitmaps scan the input arrays word by word, even when most bits are 0.

Can we keep the useful operations while using less space?

Illustrative calculation. Decimal MB/GB. Excludes metadata and other index overhead.

Bitmap patterns

Example	32 bits			
A	00000000	00000100	00000000	00000000
B	00000000	00000000	11111111	11111111
C	00110011	00110011	00110011	00110011

A short thought experiment

How would you describe each bitmap without listing every bit?

Compression strategies

Few set bits

Store the positions of the 1s.

Example A:
position 14.

Long runs

Store a bit value and a run length.

Example B:
16 zeros, 16 ones.

Repeated patterns

Describe a pattern and reuse it.

Example C:
0011, eight times.

Representation trade-offs

Each idea has its own overhead and supported operations.

Conceptual descriptions, not complete binary encodings. Positions start at 1.

Compression and query processing

Storage question

How many bytes does the representation need?

Query question

How much work does an AND or OR operation require?

- Some representations support operations directly on compressed data.
- Operations on compressed data can read fewer bytes from memory and skip uniform regions.
- Decoding and bookkeeping can also add work.

The smallest representation need not give the fastest query.

Planned course structure

1 Compressed Bitmap Indexes

Assignment 1: Compression Benchmark

2 Decision Diagrams

Assignment 2: Variable Ordering Explorer

3 Modern Compressed and Succinct Data Structures

Assignment 3: Paper Reproduction

Course organisation

Meetings

- **Schedule:**

Kickoff: 6 Oct 2026

Tuesdays

13 Oct–17 Nov 2026: 11:45–14:00

24 Nov 2026–26 Jan 2027:

12:00–14:15

- **Location:**

Until 17 Nov: Hörsaal I –

Christian Doppler (JAK2AUG0.70)

From 24 Nov: Hörsaal II –

Lise Meitner (JAK2AUG0.69)

Materials and contact

- **Course platform:**

[ATDA course website](#)

annbolotina.github.io/teaching/ATDA/

- **Email:**

anna.bolotina@plus.ac.at

- **Optional consultations:**

27 Oct 2026, 11:45–14:00

19 Jan 2027, 12:00–14:15

Before report deadlines.

Attendance is not compulsory.

No meetings: 3 Nov and 8 Dec 2026; Christmas break: 19 Dec 2026–6 Jan 2027. [PLUSonline](#)

Module 1: Compressed Bitmap Indexes

Topics

- Bitmap indexes and Boolean queries
- Density and runs
- WAH, EWAH and Roaring
- Space and query time

Compression Benchmark

Compare bitmap representations on data with different patterns.

Measure storage and AND/OR time.
Explain the differences.

How do bitmap representations compare in memory use and query performance?

Module 2: Decision Diagrams

A different way to represent the same answer

A graph of yes/no decisions can reuse common subproblems.

Topics

- Boolean functions and BDDs
- Reduced, ordered diagrams
- Variable ordering

Variable Ordering Explorer

Use an existing BDD library. Change the order of decisions and measure the diagram size.

Can the same function have very different representation sizes?

Module 3: Modern Compressed and Succinct Structures

Topics

- Succinct data structures
- Tree-encoded bitmaps (TEB)
- Recursive sharing: CFLOBDDs
- Selected research papers

Paper Reproduction

Choose a paper and reproduce one manageable experiment, figure or table.

Compare your observations with the authors' results.

How do we test a claim from a research paper?

Design of a bitmap experiment

Example question

Does clustering change query time when the number of 1s stays fixed?

- 1 Generate bitmaps with equal lengths and numbers of 1s.
- 2 Vary how the 1s cluster, and define the query operands.
- 3 Check correctness, then measure storage and operation time.
- 4 Explain the result and the limits of the experiment.

Learning outcomes

By the end of the track, you should be able to:

- Explain how data patterns affect compressed representations.
- Compare memory use and query performance fairly.
- Investigate a structural choice such as variable ordering.
- Read a paper and evaluate one of its experimental claims.

Starting point

Useful background

- Basic data structures
- Programming experience
- Simple Boolean logic

What we introduce in the track

- Bitmap compression
- Decision diagrams
- Succinct representations

Prior knowledge of compressed data structures or decision diagrams is **not required**.

Assignment submissions

Assignment	Report due	Presentation
1. Compression Benchmark	3 Nov 2026	10 Nov 2026
2. Variable Ordering Explorer	24 Nov 2026	1 Dec 2026
3. Paper Reproduction	19 Jan 2027	26 Jan 2027

Individual or group work will be decided once enrolment is final, before the first assignment is released.

All reports are due by **23:59** on the listed date (Salzburg local time).

Submission requirements

- **Required deliverables:** Written report and presentation.
- **Report submission:** Upload to Blackboard.

Assessment

Assignment	Presentation	Report	Total
Compression Benchmark	20%	10%	30%
Variable Ordering Explorer	20%	10%	30%
Paper Reproduction	30%	10%	40%

- **Evaluation criteria:** Clear explanation of methods and results, and answering questions.
- **Late submissions:**
Extensions may be granted for illness or other exceptional circumstances.
Contact me as soon as possible.

Participation and collaboration

- **Attendance**

Attendance is required; up to two absences are allowed.
Optional consultations do not count towards this limit.

- **Collaboration between students**

Discussing ideas and methods is encouraged.
Your report and presentation must reflect your own work
and understanding.

Use of sources and AI tools

Sources and reused material

Cite papers, datasets, libraries and any reused text or figures.

AI tools

- **Permitted use:** AI may assist with brainstorming, programming and language editing.
- **Disclosure:** Briefly state which tools you used and for what purpose in your report.
- **Responsibility:** Verify generated content and references. Your work must reflect your own understanding, and you must be able to explain your methods and results.

[PLUS guide to AI tools in assessed work \(September 2025, in German\)](#)

Suggested AI assistant

Antigravity CLI

- Use it in Terminal.
- Free tier; no subscription required.
- Recommended model: **Gemini Flash**.
- A starter framework will be provided for the assignments.
- Usage limits apply. Check model quotas with `/usage`.

[Antigravity CLI documentation](#) [Models and availability](#)

Upcoming sessions

1 **Week 2: Introduction to Bitmap Indexes**

Bitmap indexes and Boolean queries.

Run-length encoding; the effects of density and clustering.

2 **Week 3: Compressed Bitmap Representations**

WAH, EWAH and Roaring: core ideas and trade-offs.

Compare formats on different data patterns.

Assignment 1: requirements, metrics and correctness checks.

Your interests and experience

- What interested you in this track?
- Have you worked with database indexes or compression before?
- Which programming languages are you comfortable using?

Questions?

Anna Bolotina

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Density alone does not describe a bitmap

Pattern	32 bits			
Long runs	00000000	00000000	11111111	11111111
Repetition	00110011	00110011	00110011	00110011

Both have 16 ones in 32 positions: **50% density**.

Different patterns, same density

The lengths of runs and the repeated subpatterns differ.